SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**

Question Count: 5

Code of Conduct

I G. Yashwita (192511032) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Yashwita

Assessment Tasks

Scenario-Based Questions

SET 3 :

1. A manufacturing company plans to migrate its legacy enterprise applications and databases from an on-premises data center to the cloud. The organization wants to minimize downtime while ensuring application performance and security.

- Design a cloud migration strategy suitable for the organization.
- Explain how different migration approaches (Rehost, Refactor, Replatform, Repurchase, Retire, Retain) can be applied.
- Discuss methods to validate application performance after migration.
- Recommend strategies to reduce migration risks and downtime.

2. A smart city project deploys thousands of IoT sensors to monitor traffic, air quality, energy consumption, and public safety. The collected data must be processed and analyzed in real time for decision-making.

- Design a cloud-based architecture for IoT data collection and analytics.
- Explain how message brokers, stream processing, and cloud storage are integrated.
- Discuss methods for handling high-volume, high-velocity sensor data.
- Recommend techniques to ensure scalability, reliability, and security.

3. A global e-commerce company wants executives to monitor sales performance, customer behavior, and regional trends through interactive dashboards. The solution must support real-time analytics and historical reporting.

- Design a cloud-based business intelligence architecture.
- Explain how data visualization tools integrate with cloud data warehouses.
- Discuss techniques for optimizing dashboard performance with large datasets.
- Recommend security measures to protect business reports and sensitive data.

4. An online gaming company experiences fluctuating workloads throughout the day. During peak hours, users report high latency and reduced application performance.

- Design a cloud monitoring and performance optimization solution.
- Explain how cloud monitoring tools, metrics, logs, and tracing help identify performance bottlenecks.
- Discuss strategies for resource optimization and capacity planning.
- Recommend methods to improve application availability while controlling operational costs.

5. A financial institution stores petabytes of customer transaction data for regulatory compliance and business analytics. The organization must retain historical records while managing storage costs efficiently.

- Design a cloud-based data lifecycle management strategy.
- Explain how data classification, retention policies, and archival storage should be implemented.
- Discuss the importance of governance, auditing, and regulatory compliance in big data environments.
- Recommend techniques for balancing data accessibility, security, and cost optimization.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1. Cloud Migration Strategy for a Manufacturing Company:

Cloud migration is the process migrate its legacy applications and databases to the cloud using a phased migration strategy. First, the six migration approaches are Rehost, Refactor, Replatform, Repurchase, Retire and Retain.

After migration, load testing, response-time measurements and database performance checks should be conducted. Monitoring CPU, memory and network usage helps identify bottlenecks. Regular backups, data replication, encryption and a rollback plan, reduce migration risks. Migrating during low-production hours and moving applications in stages helps minimize downtime and maintain business continuity.

First, it should access existing applications, dependencies, performance requirements and security risks. A pilot migrations can be performed before moving critical systems.

Q. Cloud-Based IOT Architecture for a Smart City:

A smart city can use a cloud-based IOT architecture to collect and analyze data from traffic, air quality, energy and public safety sensors. Sensors transmit data through IoT gateways using protocols such as MQTT or HTTPS. Cloud IoT services authenticate devices and forward data to message brokers.

Message brokers such as Apache Kafka distribute incoming events to stream processing services for real-time analysis. Processed data is stored for cloud databases for immediate access and object storage for historical analysis. Edge computing, filtering and data compression

Scalability can be achieve through automatic scaling and data partitioning. Reliability requires redundant services, backups and replication. Security measures include device authentication, encryption, access control and regular updates. Dashboards and automated alerts help authorities respond quickly to

traffic congestion, pollution and other emergencies.

3. Cloud - Based Business Intelligence for E-commerce:

A global e-commerce company can implement cloud-based Business Intelligence (CBI) to analyze sales, customer behaviour and regional trends. Data from online stores, payment systems and inventory database is collected through ETL/ELT pipelines and stored in a cloud data warehouse.

BI tools such as Power BI and Tableau connect to the warehouse to create interactive dashboards and reports. Executives can monitor revenue, product performance, customer preferences and regional sales through real time or scheduled updates.

Dashboard performance can be improve using caching, Data aggregation, partitioning, optimized queris and incremental data loading.

Security can be maintained through role-based access control, encryption, multi-factor authentication and audit logs. Row-level security ensures that users access only authorized information.

4. Cloud monitoring and Performance Optimization for Online Gaming :

An online gaming company can improve performance by deploying game services across multiple availability zones and distributing player requests through load balancers. Auto-scaling increases resources during peak hours and reduces them during low-demand periods. Caching and database optimization further reduce latency.

Cloud monitoring tools collect metrics such as CPU usage, memory consumption, response time and network latency. Logs record errors and system events while distributed tracing identifies delays across application services.

Resource optimization involves right-sizing virtual machines, conducting load tests and analyzing historical workloads for capacity planning. Redundant services, health checks, and automated failover improve availability. Cost monitoring and scaling down unused resources reduce expenses.

5- **Cloud-Based Data Lifecycle Management for a Financial Institution:**

A financial institution can manage petabytes of transaction data using a cloud-based data lifecycle strategy. Data should be classified according to sensitivity, access frequency and regulatory requirements. Frequently accessed records can remain in high-performance storage, while older records can be transferred to lower-cost or archival storage.

Automated lifecycle policies should define retention periods and

move data between storage tiers. Records must be deleted only after the required retention period ends and applicable legal holds are cleared. Versioning and immutable storage can protect important records from accidental deletion or modification.

Data governance, audit logs, encryption and role-based access control help ensure security and regulatory compliance. Multi-factor authentication restricts access to authorized users. Compression and automated storage tiering reduce costs while retrieval policies maintain access to archived records when required. This strategy balances data security, accessibility, compliance and storage efficiency.